

The relative p -Wasserstein distance is a metric and its two coupling formulas agree

Candidate solution packet for arXiv:2411.05678

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Status. Candidate full solution, likely valid, pending human review. For every $1 \leq p < \infty$, the packet proves $\widehat{W}_p = W_p$ and proves identity of indiscernibles for W_p . Thus it answers both questions posed for $p > 1$ in Section 6.3 of the source paper.

1 Source questions

The source is P. Bubenik and A. Elchesen, *Relative Optimal Transport*, arXiv:2411.05678v3 (6 April 2026) [1]. In Section 6.3, on source PDF pages 31 and 32, the authors ask:

“It is an open question whether $\widehat{W}_p = W_p$ for $p > 1$.”

“It is an open question whether $W_p(\mu, \nu) = 0$ implies that $\mu = \nu$ when $p > 1$.”

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Figure 1: The two questions on source PDF pages 31 and 32.

2 Definitions and notation

Let (X, d, A) be a metric pair as in the source: (X, d) is complete and separable, and $A \subset X$ is nonempty and closed. Put

$$d_A(x) := d(x, A), \quad d_p(x, y) := \min \left\{ d(x, y), (d_A(x)^p + d_A(y)^p)^{1/p} \right\}.$$

A relative Borel measure is a Borel measure modulo measures supported on A . Let $\mathcal{M}_p^+(X, A)$ be the source’s space of tight relative measures μ with

$$\mu(d_A^p) = \int_{X \setminus A} d_A(x)^p d\mu(x) < \infty.$$

For $\delta > 0$, write μ_δ for the restriction to $\{d_A > \delta\}$ and μ^δ for the restriction to $\{0 < d_A \leq \delta\}$. The source proves

$$\mu^\delta(d_A^p) \longrightarrow 0 \quad \text{as } \delta \downarrow 0. \quad (1)$$

Let $\Pi(\mu, \nu)$ be the set of relative couplings. Fix $a \in A$. The structured class $\Pi_\delta(\mu, \nu)$ consists of the trivial extensions

$$\rho = \sigma + (i_1^a)_* \mu^\delta + (i_2^a)_* \nu^\delta, \quad \sigma \in \Pi(\mu_\delta, \nu_\delta),$$

where $i_1^a(x) = (x, a)$ and $i_2^a(y) = (a, y)$. The two quantities in the first open question are

$$\begin{aligned} \widehat{W}_p(\mu, \nu)^p &:= \inf_{\pi \in \Pi(\mu, \nu)} \int d_p(x, y)^p d\pi(x, y), \\ W_p(\mu, \nu)^p &:= \inf_{\delta > 0} \inf_{\rho \in \Pi_\delta(\mu, \nu)} \int d_p(x, y)^p d\rho(x, y). \end{aligned}$$

Since every structured coupling is a coupling, the source observes that $\widehat{W}_p \leq W_p$.

3 Proof intuition

The first question is resolved by approximating an *arbitrary* coupling, not an optimizer. Collapse only the δ -neighborhood of the reservoir A onto one point of A , and then restore the two collapsed marginal tails by the trivial reservoir couplings. The resulting coupling is structured. The collapsed integrand converges pointwise to the original integrand and is dominated by $d_A(x)^p + d_A(y)^p$, whose coupling integral is exactly the sum of the two finite marginal p -moments. Dominated convergence, together with (1), therefore preserves the cost in the limit. This gives the missing inequality $W_p \leq \widehat{W}_p$ without needing a quantitative $p > 1$ retraction estimate.

For definiteness, take a compact set K away from A . A particle starting in K and ending outside a small ordinary d -neighborhood K_r must pay at least r in the relative metric d_p : both the direct branch and the reservoir branch in the definition of d_p exceed r . A coupling of arbitrarily small cost therefore gives $\mu(K) \leq \nu(K_r)$. The set K_r has finite ν -mass because it remains a positive distance from A . Continuity from above as $r \downarrow 0$, followed by symmetry and tightness, then forces $\mu = \nu$. No properness or compactness of K_r is used.

4 The full result

Theorem 1. *Let (X, d, A) be a complete separable metric pair with $A \neq \emptyset$, and let $1 \leq p < \infty$. For all $\mu, \nu \in \mathcal{M}_p^+(X, A)$,*

$$W_p(\mu, \nu) = \widehat{W}_p(\mu, \nu).$$

Moreover,

$$W_p(\mu, \nu) = 0 \implies \mu = \nu.$$

Consequently the pseudometric W_p constructed in the source is a metric on $\mathcal{M}_p^+(X, A)$. In particular, both open questions stated there for $p > 1$ have affirmative answers.

Proof. We first prove equality of the two costs. Fix a coupling $\pi \in \Pi(\mu, \nu)$. For $\delta > 0$, define the Borel map

$$q_\delta(x) := \begin{cases} a, & 0 < d_A(x) \leq \delta, \\ x, & \text{otherwise,} \end{cases} \quad R_\delta(x, y) := (q_\delta(x), q_\delta(y)).$$

This is the retraction used in Section 6.2 of the source. Its coupling lemma gives

$$\sigma_\delta := (R_\delta)_* \pi \in \Pi(\mu_\delta, \nu_\delta).$$

Trivially extend it to

$$\rho_\delta := \sigma_\delta + (i_1^a)_* \mu^\delta + (i_2^a)_* \nu^\delta \in \Pi_\delta(\mu, \nu).$$

Because $d_p(x, a) = d_A(x)$, its cost is exactly

$$\begin{aligned} \int d_p^p d\rho_\delta &= \int_{X^2} d_p(q_\delta(x), q_\delta(y))^p d\pi(x, y) \\ &\quad + \mu^\delta(d_A^p) + \nu^\delta(d_A^p). \end{aligned} \tag{2}$$

For every $x \in X$, $q_\delta(x) \rightarrow x$ as $\delta \downarrow 0$. Continuity of d_p therefore gives pointwise convergence of the integrand in (2). Furthermore,

$$\begin{aligned} d_p(q_\delta(x), q_\delta(y))^p &\leq d_A(q_\delta(x))^p + d_A(q_\delta(y))^p \\ &\leq d_A(x)^p + d_A(y)^p. \end{aligned} \tag{3}$$

The last function is π -integrable, since the coupling marginals give

$$\int_{X^2} (d_A(x)^p + d_A(y)^p) d\pi(x, y) = \mu(d_A^p) + \nu(d_A^p) < \infty. \tag{4}$$

Dominated convergence and (1) applied to (2) now yield

$$\lim_{\delta \downarrow 0} \int d_p^p d\rho_\delta = \int d_p^p d\pi.$$

Since every ρ_δ is admissible in the definition of W_p ,

$$W_p(\mu, \nu)^p \leq \int d_p^p d\pi.$$

Taking the infimum over arbitrary $\pi \in \Pi(\mu, \nu)$ gives $W_p \leq \widehat{W}_p$. Together with the reverse inequality already noted in the source, this proves $W_p = \widehat{W}_p$.

We next prove identity of indiscernibles directly. Assume $W_p(\mu, \nu) = 0$. Let $K \subset X \setminus A$ be compact and put

$$\alpha := d(K, A) > 0.$$

Fix $0 < r < \alpha$ and let

$$K_r := \{y \in X : d(y, K) \leq r\}.$$

The function d_A is 1-Lipschitz, so $d_A(y) \geq \alpha - r$ on K_r . Hence

$$\nu(K_r) \leq (\alpha - r)^{-p} \nu(d_A^p) < \infty. \tag{5}$$

For $x \in K$ and $y \notin K_r$, one has $d(x, y) > r$, while

$$(d_A(x)^p + d_A(y)^p)^{1/p} \geq d_A(x) \geq \alpha > r.$$

Thus both branches defining $d_p(x, y)$ exceed r , and consequently

$$d_p(x, y) > r \quad (x \in K, y \notin K_r). \tag{6}$$

Given $\eta > 0$, the definition of $W_p = 0$ supplies some $\delta > 0$ and $\rho \in \Pi_\delta(\mu, \nu)$ such that $\int d_p^p d\rho < \eta$. Marginality, (6), and Markov's inequality give

$$\begin{aligned}\mu(K) &= \rho(K \times X) \leq \rho(K \times K_r) + \rho(K \times (X \setminus K_r)) \\ &\leq \nu(K_r) + r^{-p} \int d_p^p d\rho \\ &< \nu(K_r) + r^{-p} \eta.\end{aligned}$$

Letting $\eta \downarrow 0$ gives $\mu(K) \leq \nu(K_r)$. Choose any decreasing sequence $r_n \downarrow 0$ with $r_1 < \alpha$. Then $K_{r_n} \downarrow K$, and (5) makes the first set finite. Continuity from above therefore yields

$$\mu(K) \leq \nu(K).$$

Interchanging μ and ν gives equality on every compact $K \subset X \setminus A$. Both measures are tight by the definition of $\mathcal{M}_p^+(X, A)$. The source's compact-set uniqueness lemma for tight relative measures now implies $\mu = \nu$.

The source already proves reflexivity, symmetry, and the triangle inequality for W_p on $\mathcal{M}_p^+(X, A)$. Adding identity of indiscernibles completes the metric assertion. $\square$

5 Verification report

The proof is analytic and no numerical evidence is used. The following points were checked against the definitions and lemmas in the current source:

- The retraction is Borel and its pushforward has marginals μ_δ, ν_δ (source Lemma 6.21).
- Formula (2) is exact: each tail is coupled to a point of A , where $d_p(x, a)^p = d_A(x)^p$.
- The dominator in (3) is integrable even when the coupling has infinite total mass, because (4) depends only on the two finite p -moments.
- In the definiteness proof, K_r need not be compact. Its finite mass follows from (5), which is sufficient for continuity from above.
- All marginal identities are evaluated on sets disjoint from A , so they are independent of the representative chosen for a relative measure.
- Agreement on compact subsets of $X \setminus A$ implies equality by the source's Lemma 4.1 and the tightness assumption.

The main human-review focus should be the passage from a relative coupling to the exact integral identity (2), including representative-independence, and the compact-set limiting argument following (6). Current verdict: **full solution likely valid**.

6 Scope and novelty check

The theorem uses exactly the source hypotheses: a complete separable metric pair, a nonempty closed reservoir, and tight relative measures with finite p -moment. It does not require an optimal coupling, finite total mass, properness of X , or compactness of metric neighborhoods. Although the source questions are stated for $p > 1$, the argument works uniformly for every $1 \leq p < \infty$; for $p = 1$ it recovers conclusions already proved in the source.

Before promotion, the local registry, solution, attempt, and proof-gap indexes were searched for arXiv:2411.05678 and the core phrases “relative optimal transport”, “relative Wasserstein”, “ $\widehat{W}_p = W_p$ ”, and “identity of indiscernibles”; no prior result was found. A bounded web/arXiv search on 21 July 2026 used the title, authors, exact open-question phrases, and nearby relative Wasserstein terminology. It found the source and adjacent optimal transport work but no later settlement. The current source version, v3 of 6 April 2026, still labels both questions open. This supports, but does not establish, novelty; the search was not an exhaustive review of all published literature.

References

- [1] P. Bubenik and A. Elchesen, *Relative Optimal Transport*, arXiv:2411.05678v3, 2026.